RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

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Year / Semester :	III Year / V Semester
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Academic Year :	2025 – 2026

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EXP.NO:01

STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

Aim:

To study the concept of computer forensics, its importance, and the different tools used for forensic investigation**.**

Procedure:

1. Study the fundamentals of computer forensics and understand its significance in the investigation of cybercrimes.
2. Learn the key phases of a computer forensic investigation, including identification, acquisition, preservation, examination, analysis, and documentation of digital evidence.
3. Recognize various sources of digital evidence such as computers, hard drives, USB storage devices, emails, and system log files.
4. Explore the different software tools and techniques used in computer forensic investigations.
5. Understand the purpose, functions, and practical applications of each forensic tool.
6. Analyze and compare forensic tools based on their features, capabilities, and intended use.
7. Document the findings and explain the importance of computer forensics in supporting digital crime investigations.

Introduction

Computer forensics, also known as digital forensics, is the process of identifying, collecting, preserving, examining, and analyzing digital evidence from computers and other storage media. It plays a vital role in investigating cyber incidents by helping forensic experts recover evidence, trace malicious activities, and reconstruct digital events in a legally acceptable manner.

The following tools are commonly used in forensic investigations:

S.No	Forensic Tool	Purpose
1	Autopsy	Analyzes hard disks, files, deleted files, and digital evidence.
2	FTK Imager	Creates forensic images and previews digital evidence.
3	Encase Forensic	Supports the acquisition, examination, and analysis of digital evidence.
4	Wireshark	Captures and analyzes network traffic.
5	Volatility	Analyzes memory dumps to investigate running processes and other memory artifacts.
6	Sleuth Kit	Provides command-line tools for examining disk images and file systems.
7	Magnet AXIOM	Helps examine digital evidence from computers, mobile devices, and other sources.

Computer forensic tools are designed for different tasks. Some are specialized in analyzing storage devices, while others focus on memory forensics, network traffic analysis, or acquiring digital evidence. The choice of tool depends on the nature of the investigation and the type of digital evidence being examined.

Result:

Thus, the fundamentals of computer forensics and the various forensic tools used in digital investigations were explored. The significance of collecting, preserving, examining, and analyzing digital evidence for identifying and investigating cybercrimes was understood.

2. LIVE FORENSIC INVESTIGATION USING AUTOPSY

Aim:

To conduct a basic forensic examination of a Windows system using Autopsy and identify digital evidence such as user activity, browser history, deleted files, system information, and other forensic artifacts.

Procedure:

1. Install Autopsy
Download the latest version of Autopsy from its official website and complete the installation by following the provided setup instructions.

2. Create a New Case
Open Autopsy and select New Case. Specify the case name, choose the storage location for the case, and set the report output directory.

3. Provide Case Information
Enter the required case details, including the case number, examiner's name, and any additional information. Click Finish to create the forensic case.

4. Import the Evidence
Click Add Data Source and select the appropriate evidence type, such as a disk image, local drive, or logical files. Browse to the evidence location and import the required data source.

5. Configure Analysis Modules
Choose the required Ingest Modules to perform the investigation. These modules can recover and analyze information such as files, operating system details, user accounts, browser history, downloads, cookies, email addresses, and other forensic artifacts.

6. Begin the Analysis
Click Finish to start processing the selected evidence. Autopsy scans the data source and stores the extracted information within the case database for analysis.

7. Analyze the Results
After the processing is complete, examine the recovered artifacts available in Autopsy, including:

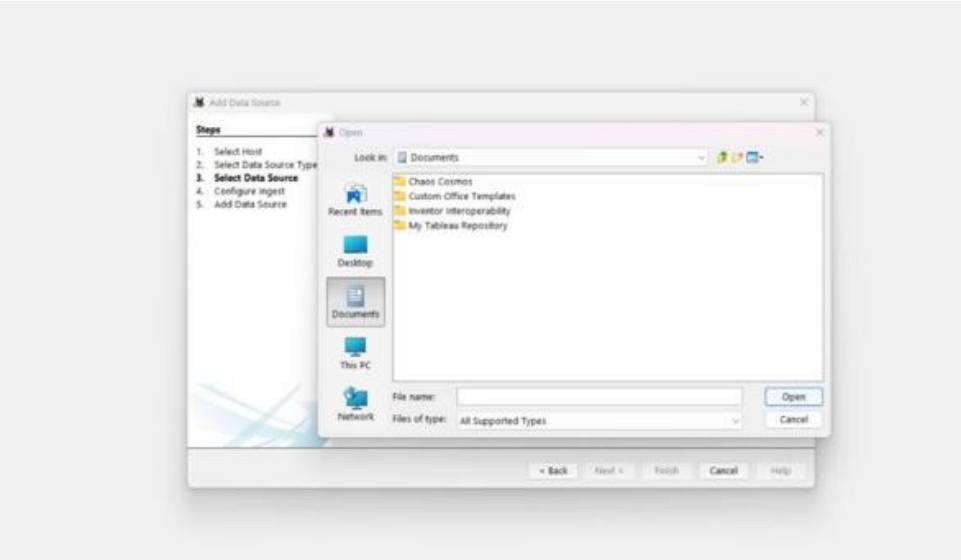
- **Connected Devices:** Review information about external storage devices connected to the system.
- **EXIF Metadata:** Inspect metadata stored in image files, including timestamps and camera information.
- **Installed Applications**: Identify software installed on the examined system**.**
- **Operating System Details:** Review information about the installed operating system.
- **User Accounts:** Examine user account information recovered from the system.
- **Recent Documents:** View records of recently accessed files and documents.
- **Browser Bookmarks and History:** Analyze browsing history and saved bookmarks**.**
- **Downloads and Search History:** Review downloaded files and browser search activities, where available.

- **Browser Cookies:** Examine cookies recovered from supported web browsers.
- **Email Addresses:** Identify email addresses extracted during artifact analysis.

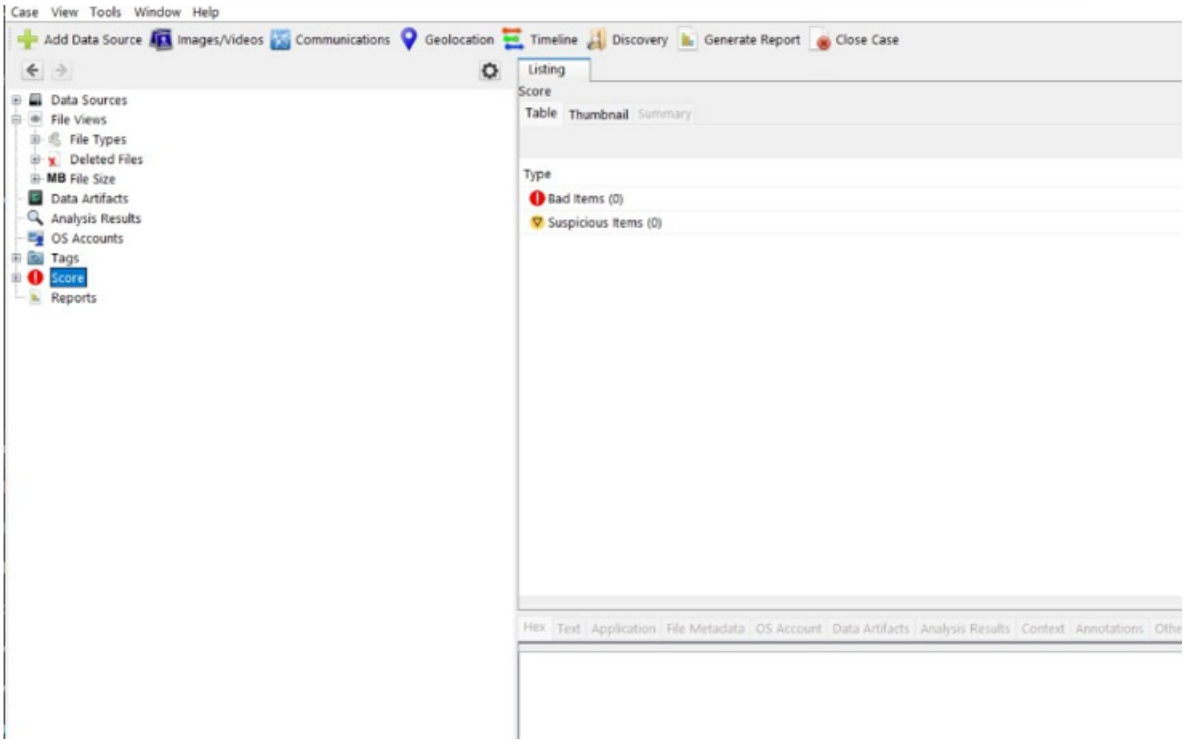
8. Create the Investigation Report

Review the collected evidence, verify the findings, and generate a forensic investigation **report** using Autopsy's reporting feature. Save the report for documentation and future reference.

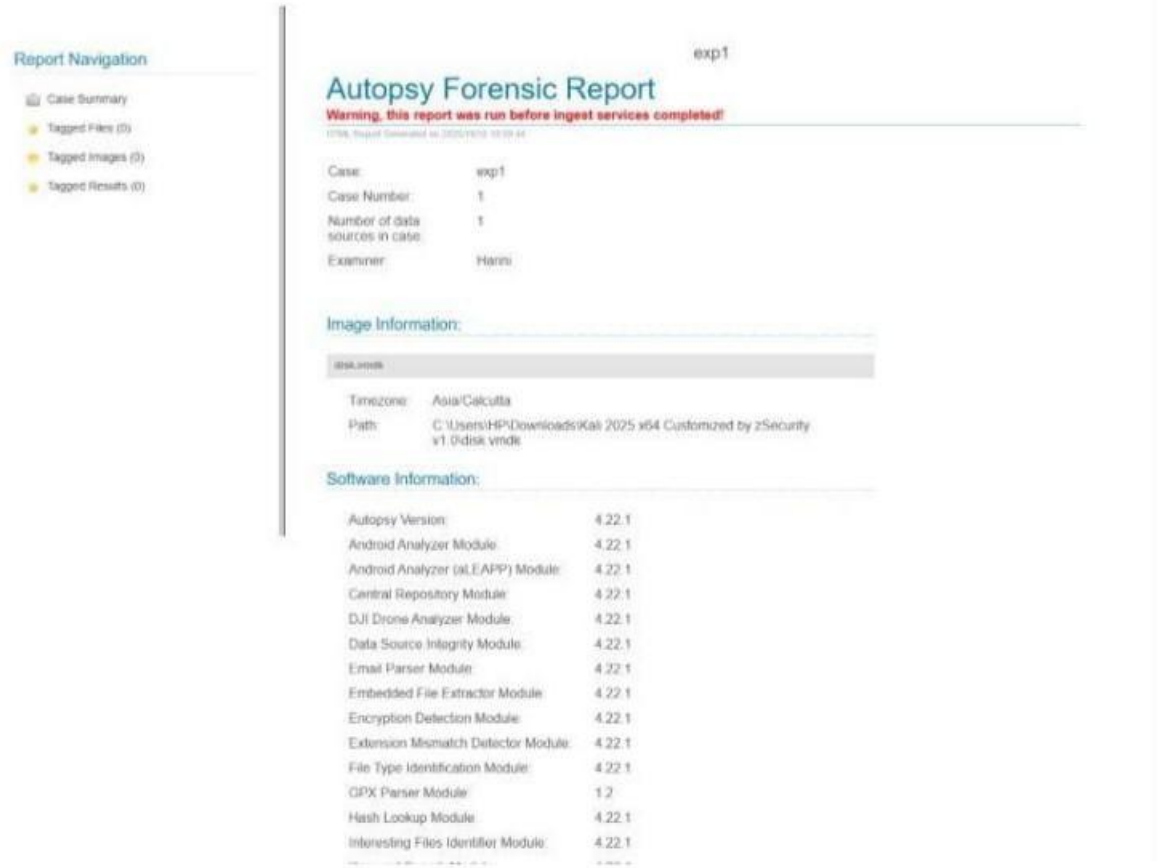
Output:



NAME: Naren S ROLL NO: 241901066



NAME: Naren S ROLL NO: 241901066



Report Navigation

- Case Summary
- Tagged Files (0)
- Tagged Images (0)
- Tagged Results (0)

exp1

Autopsy Forensic Report

Warning, this report was run before ingest services completed!

Case:	exp1
Case Number:	1
Number of data sources in case:	1
Examiner:	Hanni

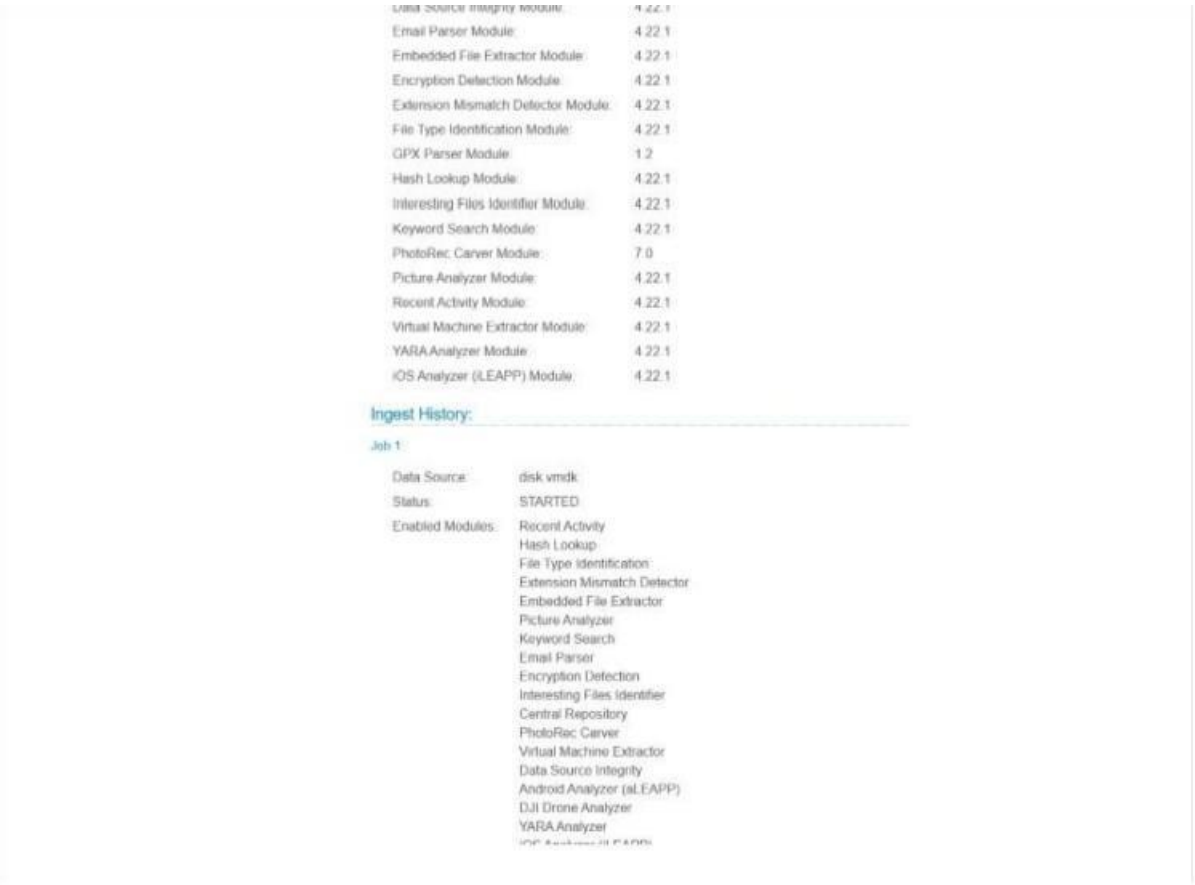
Image Information:

disk.vmdk

Timezone:	Asia/Calcutta
Path:	C:\Users\HP\Downloads\Kali 2025 x64 Customized by zSecurity v1.0\disk.vmdk

Software Information:

Autopsy Version:	4.22.1
Android Analyzer Module:	4.22.1
Android Analyzer (aLEAPP) Module:	4.22.1
Central Repository Module:	4.22.1
DJI Drone Analyzer Module:	4.22.1
Data Source Integrity Module:	4.22.1
Email Parser Module:	4.22.1
Embedded File Extractor Module:	4.22.1
Encryption Detection Module:	4.22.1
Extension Mismatch Detector Module:	4.22.1
File Type Identification Module:	4.22.1
GPX Parser Module:	1.2
Hash Lookup Module:	4.22.1
Interesting Files Identifier Module:	4.22.1



Email Parser Module	4.22.1
Embedded File Extractor Module	4.22.1
Encryption Detection Module	4.22.1
Extension Mismatch Detector Module	4.22.1
File Type Identification Module	4.22.1
GPX Parser Module	1.2
Hash Lookup Module	4.22.1
Interesting Files Identifier Module	4.22.1
Keyword Search Module	4.22.1
PhotoRec Carver Module	7.0
Picture Analyzer Module	4.22.1
Recent Activity Module	4.22.1
Virtual Machine Extractor Module	4.22.1
YARA Analyzer Module	4.22.1
iOS Analyzer (iLEAPP) Module	4.22.1

Ingest History:

Job 1

Data Source:	disk.vmdk
Status:	STARTED
Enabled Modules:	Recent Activity Hash Lookup File Type Identification Extension Mismatch Detector Embedded File Extractor Picture Analyzer Keyword Search Email Parser Encryption Detection Interesting Files Identifier Central Repository PhotoRec Carver Virtual Machine Extractor Data Source Integrity Android Analyzer (aLEAPP) DJI Drone Analyzer YARA Analyzer



Result

The Windows test system was successfully investigated using **Autopsy**. Relevant digital evidence including user files, browser activity, deleted files and file metadata was identified, examined and documented. A forensic HTML report was successfully generated.

EXP.NO: 3 FTK IMAGER – DISK IMAGING AND EVIDENCE ACQUISITION

Aim

To understand and perform disk imaging and digital evidence acquisition using FTK Imager while ensuring the integrity and authenticity of the original evidence.

Procedure

1. Download and install FTK Imager from the official Exterro (AccessData) website if it is not already available on the system.
2. Launch FTK Imager with the required administrative privileges.
3. Connect the storage device or select a valid forensic image file for analysis.
4. From the File menu, select Create Disk Image to start the imaging process.
5. Choose the appropriate source type, such as Physical Drive or Logical Drive, based on the evidence being acquired.
6. Select the desired source device and verify its information before proceeding.
7. Choose the required image format, such as E01 or Raw (dd).
8. Enter the necessary case information, including the case number, evidence number, examiner's name, and other relevant details.
9. Specify a separate destination folder to store the forensic image. Ensure that the image is not saved on the original evidence device.
10. Start the disk imaging process and wait until the acquisition is successfully completed.
11. If available, enable the image verification option and compare the generated hash values to confirm the integrity of the acquired image.

12. Record important details such as the image file information, acquisition status, and calculated hash values for documentation.

Observation:

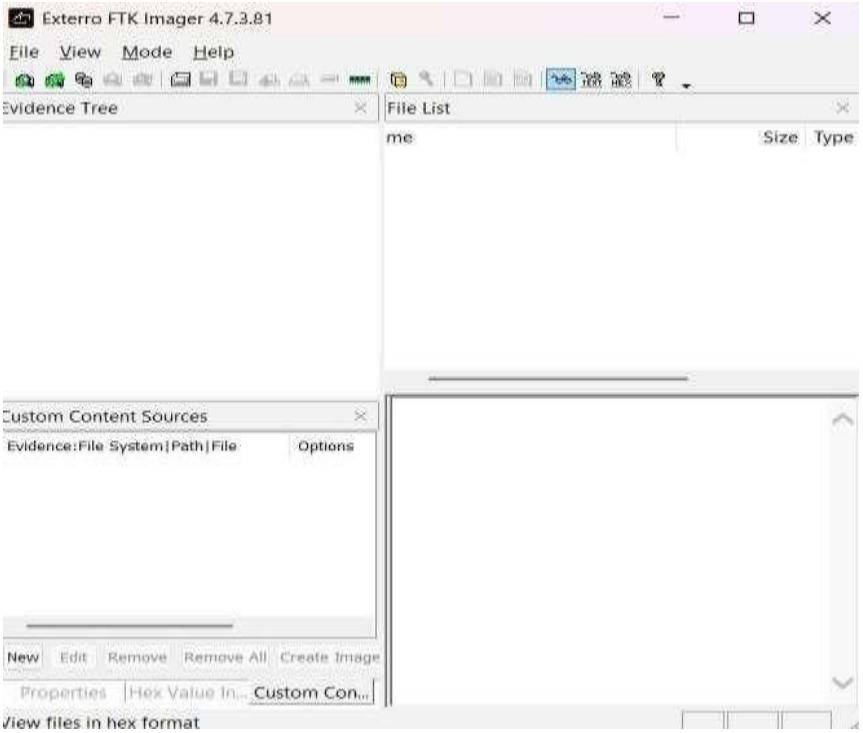
FTK Imager is a widely used digital forensic application that enables investigators to acquire and preview digital evidence while preserving the original data during a proper forensic acquisition.

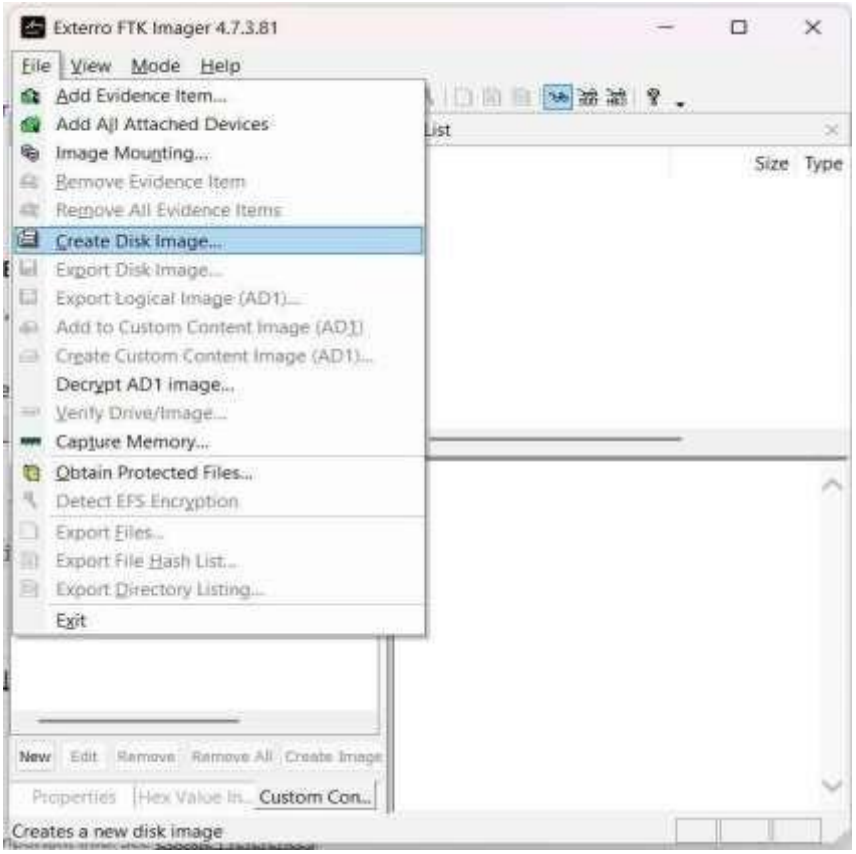
The following observations were made during the experiment:

- The source storage device and its capacity were successfully identified.
- A suitable forensic image format was selected for evidence acquisition.
- The disk imaging process generated a forensic image at the specified destination.
- The acquired image contained an exact forensic copy of the selected source data, subject to the scope of acquisition.
- Hash values, such as MD5 or SHA-1, were generated to verify the integrity of the forensic image.
- Matching hash values between the original source and the acquired image confirmed that the copied data remained unchanged during the acquisition process.

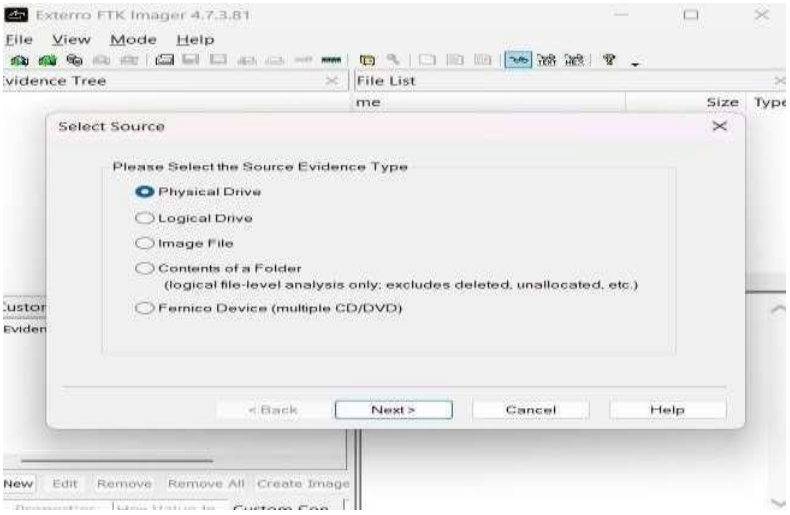
Name: NAREN S Roll no: 241901066

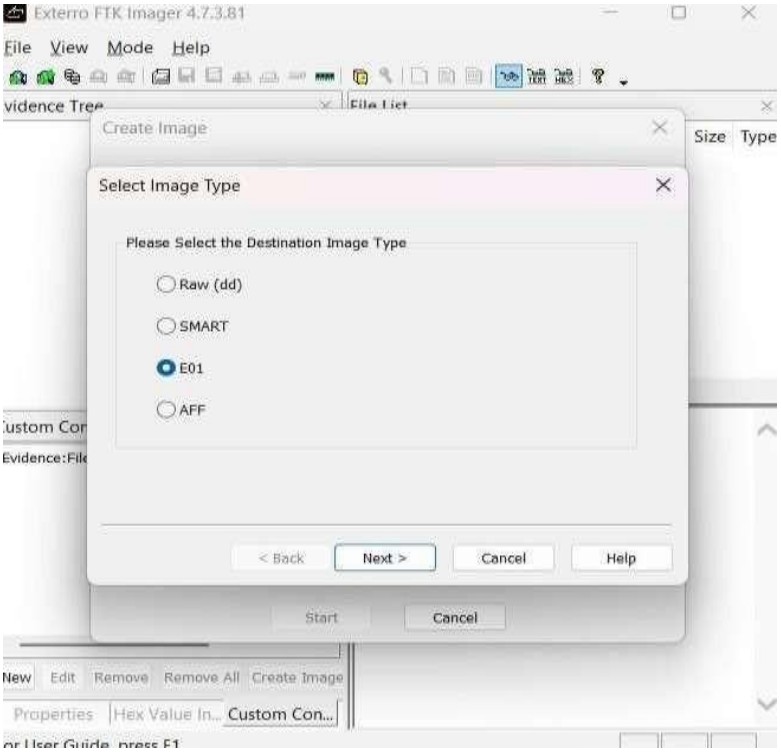
Output:

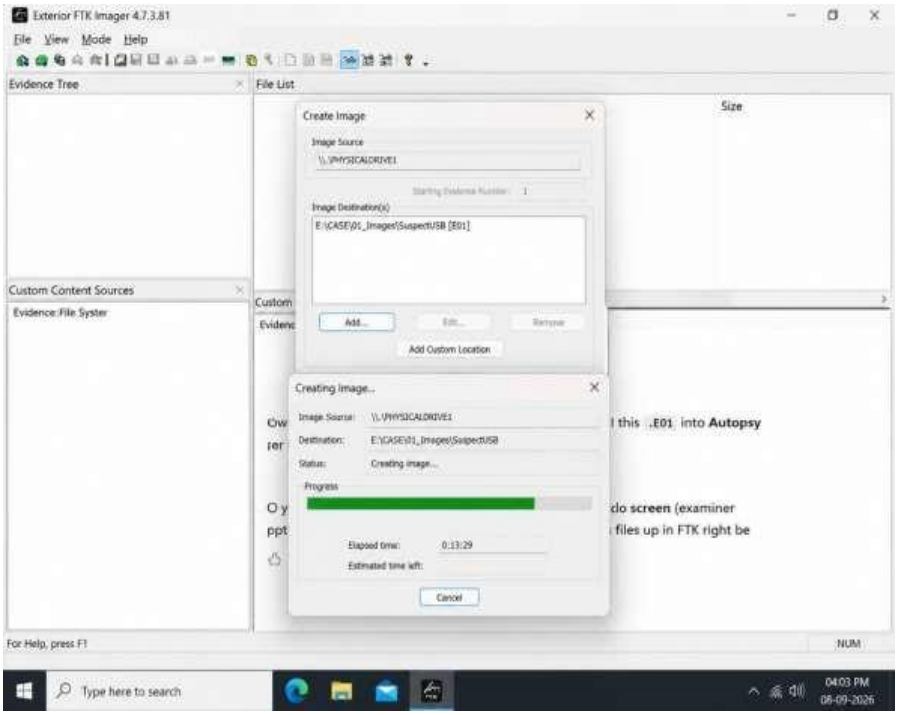




Name: NAREN S Roll no: 241901066









Result

Thus, disk imaging and digital evidence acquisition using FTK Imager were studied and performed. The process of creating a forensic image and verifying its integrity using hash values was understood.

EXP.NO: 04

USB FORENSICS – FINDING THE LAST CONNECTED USB DEVICE ON A SYSTEM

Aim:

To identify the most recently connected USB storage device on a Windows system by examining USB device history, registry entries, and related forensic artifacts.

Procedure:

1. Turn on the Windows computer and log in with an account that has administrative privileges.
2. Press **Windows + R**, type **regedit**, and press **Enter** to open the Windows Registry Editor.
3. Navigate to the following registry path to view the history of USB storage devices:

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Enum\USBSTOR

4. Expand the **USBSTOR** key and examine the available entries to identify USB storage devices that have been connected to the system.
5. Open the subkeys of each device to view information such as the device model, serial number, and instance ID.
6. Navigate to the following registry location to obtain additional USB device information:

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Enum\USB

7. Examine the registry entries to identify details such as the manufacturer, product name, vendor ID (VID), product ID (PID), device class, and serial number, if available.
8. Use a forensic registry analysis tool, such as **USB Detective**, **Registry Explorer**, or another Windows registry analysis utility, to review device properties and associated timestamps.
9. Open the Windows device installation log located at:

C:\Windows\INF\setupapi.dev.log

Review the log entries to identify USB device installation events and connection records.

10. Compare the registry information, installation logs, and available timestamps to determine the USB storage device that was most recently connected to the computer.
11. Document the identified USB device details, connection history, timestamps, and other relevant observations in the investigation record.
12. Analyze the collected information to understand the sequence of USB device usage and verify the findings using multiple forensic artifacts wherever possible.

Observation:

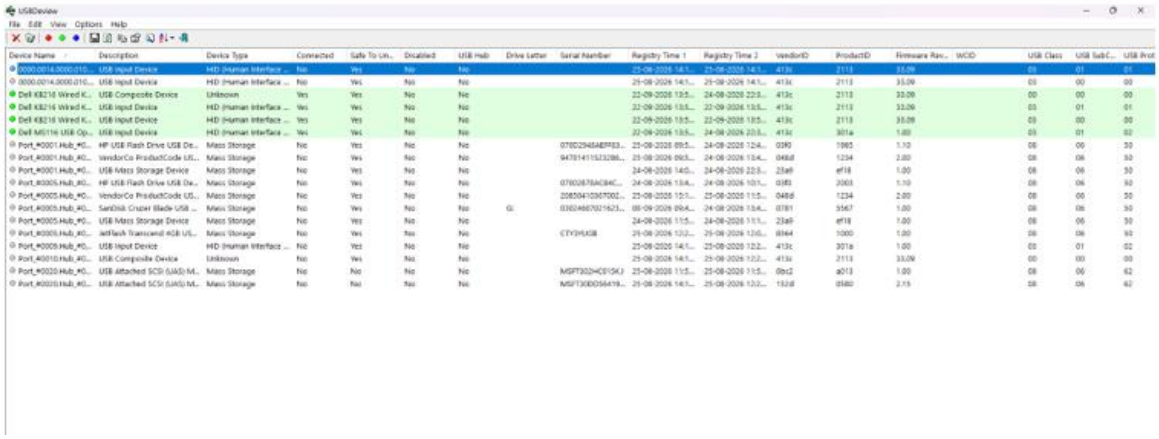
The USB device history stored in the Windows Registry provides valuable information about storage devices that have been connected to the system.

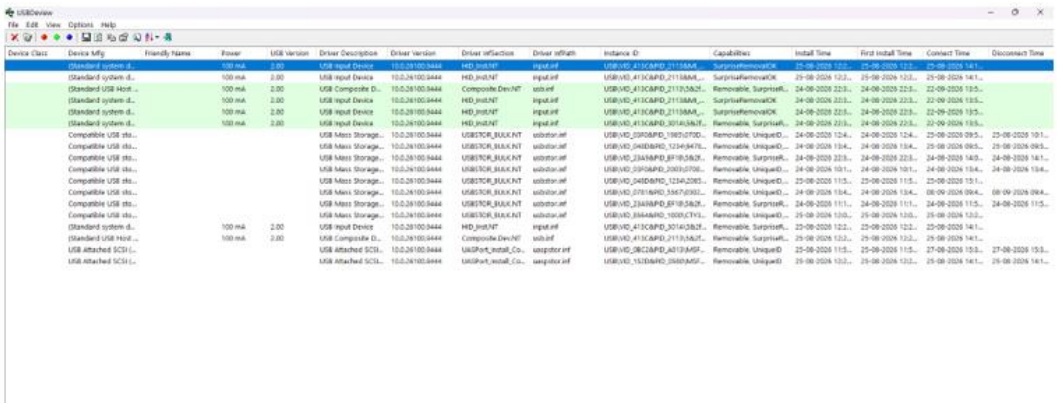
The following observations were made during the experiment:

- Previously connected USB storage devices were successfully identified from the **USBSTOR** registry key.

- Device details such as the manufacturer, model name, vendor ID, product ID, and serial number were available for several devices.
- The registry contained device instance information that helped distinguish between different USB devices.
- The Windows SetupAPI log provided records related to USB device installation and detection.
- Registry timestamps and installation logs were compared to identify the most recently connected USB storage device.
- The collected registry entries and system logs together provided useful digital evidence for reconstructing USB device activity on the Windows system.

Output:





Observation:

The Windows Registry maintains information about USB devices that have been recognized by the computer. These entries may help investigators identify previously connected USB storage devices.

The following details can be observed during the investigation:

S.No	Parameter	Observation
1	Operating System	Windows
2	Forensic Technique	USB device history analysis
3	Registry Location	USBSTOR
4	Device Information	Manufacturer and model
5	Device Identifier	Serial number or instance ID, if available
6	Connection Evidence	Registry entries and system logs
7	Last Connected Device	Record the device identified from the available evidence
8	Timestamp	Record the relevant date and time, if available

The registry and system logs can provide evidence of previously recognized USB devices. The exact connection time may not be available from every registry entry, so additional logs or forensic artifacts may be required to establish the most recent connection.

Result:

Thus, USB forensics was studied by examining Windows Registry entries and system logs to identify previously connected USB storage devices. The available device information and timestamps were analyzed to determine the most recently connected USB device.

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5. Viewing the Last Activity of a PC Using Windows Forensics

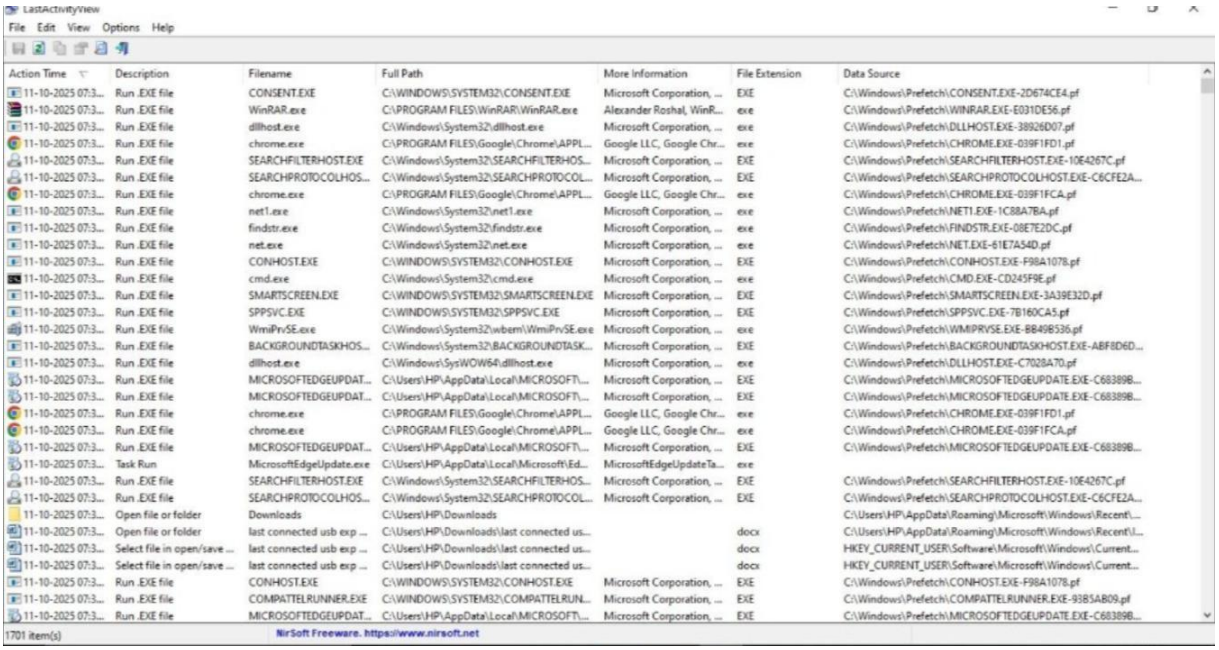
Aim:

To examine the recent activity of a Windows PC using system logs, recent files, and other Windows artifacts for digital forensic investigation.

Procedure:

1. Start the Windows computer and log in using an account with the necessary administrative privileges.
2. Open **File Explorer** and navigate to **Home** or **Quick Access** to view recently opened files and folders, if the feature is enabled.
3. Press **Windows + R**, type **recent**, and press **Enter** to open the **Recent Items** folder and examine shortcuts to recently accessed files.
4. Open the **Run** dialog again, type **eventvwr.msc**, and press **Enter** to launch the Windows Event Viewer.
5. Expand **Windows Logs** and select **System** to review system events such as startup, shutdown, hardware changes, and driver activities.
6. Open **Windows Logs → Security** to examine security audit events, including successful and failed logon attempts, user account activities, and other security-related events.
7. Navigate to **Windows Logs → Application** to inspect application events, software errors, warnings, and application execution records.
8. Review each event carefully by examining its **Event ID**, source, date, time, severity level, and detailed description to understand the recorded system activity.
9. Identify important startup and shutdown events from the **System** log to determine when the computer was powered on or off.
10. Open the web browser and review the browsing history, downloads, and recently visited websites, if access is authorized and the history is available.
11. Examine additional Windows artifacts, such as the **Recycle Bin**, **Prefetch files**, **Recent Documents**, and **Jump Lists**, if available, to gather more evidence of user activity.
12. Record all relevant observations, timestamps, event IDs, and recovered artifacts in the investigation notes for further forensic analysis.

Output:



Observation:

Windows stores different types of activity information in system logs, recent-item records, and application artifacts. These sources can help an investigator understand some of the activities performed on a computer.

S. No.	Activity Source	Observation
1	Recent Items	Recently accessed files or shortcuts, if available
2	Event Viewer – System	System startup, shutdown, and other system events
3	Event Viewer – Security	Recorded login and security events, depending on audit settings
4	Event Viewer – Application	Application-related events and errors
5	Browser History	Recently visited websites, if retained
6	Date and Time	Timestamp associated with the selected event or artifact

The available records depend on Windows settings, user activity, audit policies, and log retention. Recent-file entries and browser history may be incomplete or deleted. A recorded event should not automatically be treated as proof of a person's actions.

Result:

Thus, the recent activity of a Windows PC was examined using Windows Event Viewer, recent-file records, and browser history. The available artifacts and timestamps were studied to understand how computer activity can be investigated using digital forensic techniques.

Name: NAREN S Roll No: 241901066

EXP.NO:06

EXTRACTION OF BROWSER ARTIFACTS FOR DIGITAL FORENSIC INVESTIGATION

Aim:

To extract and examine browser artifacts such as browsing history, cookies, cached files, downloads, and saved browsing information for digital forensic investigation.

Procedure:

1. Start the computer and log in to the Windows system using an account with the required permissions.
2. Identify the web browser installed on the system, such as **Google Chrome**, **Microsoft Edge**, or any other supported browser.
3. Open the browser and access the browsing history by pressing **Ctrl + H** or by selecting the **History** option from the browser menu.
4. Examine the browsing history and record the available information, including website URLs, page titles, visit dates, and timestamps.
5. Open the browser's download history by pressing **Ctrl + J** and review the downloaded files, their download locations, and the corresponding timestamps.
6. Navigate to the browser's user profile directory. For example, Google Chrome typically stores user profile data at:
7. **%LOCALAPPDATA%\Google\Chrome\User Data\Default**
8. Locate and examine important browser artifact files, including:

   History – Stores browsing history and download records.

   Cookies – Contains website cookie information.

Web Data – Stores autofill information and other browsing-related data.

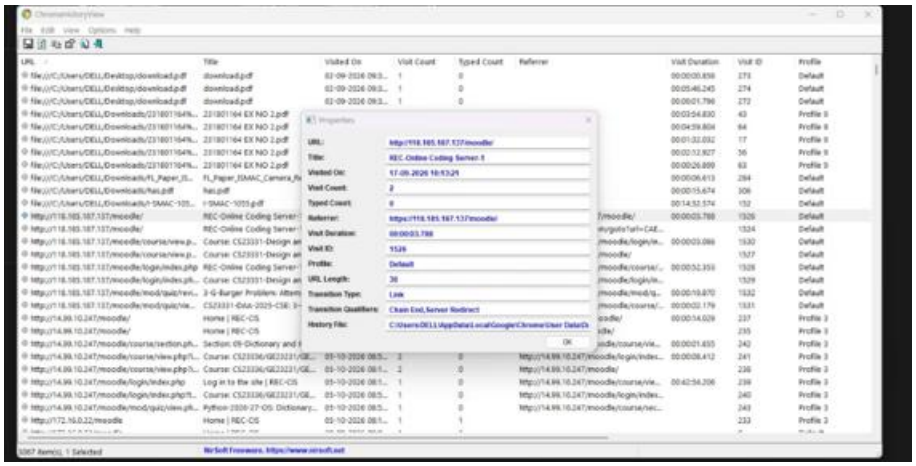
Login Data – Contains saved login credentials in encrypted form.

Cache – Stores temporary web resources such as images, scripts, and webpages.

9. Launch **Autopsy** or another authorized forensic tool and create a new case or open an existing investigation.
10. Add the forensic disk image or an authorized copy of the browser profile as the data source and allow the tool to process the available browser artifacts.
11. Review the extracted results, including visited URLs, timestamps, downloads, cookies, bookmarks, and other browser-related information available from the evidence.
12. Compare the recovered browser artifacts with the manually observed browser history, wherever applicable, to verify the findings.
13. Record all relevant observations, timestamps, recovered browser artifacts, and other investigation details in the observation table.

Precaution: Use only data you are authorized to examine. Avoid opening or sharing private passwords or sensitive personal information. For forensic work, preserve the original evidence and analyze a verified copy whenever possible.

Output:



Observation:

Browser artifacts provide useful information about a user's web activity. These artifacts can help investigators reconstruct browsing-related events, subject to the information retained by the browser.

S.No	Artifact	Information Obtained
1	Browsing History	Visited URLs, page titles, and available timestamps
2	Download History	Downloaded file names, locations, and related timestamps
3	Cookies	Website-related session and preference information
4	Cache	Cached web resources and related metadata
5	Web Data	Certain autofill and browsing-related records
6	Login Data	Stored login-related records; sensitive values should not be exposed
7	Forensic Tool	Autopsy or another suitable browser artifact analysis tool

The examination may reveal website visits, download activity, and other browsing-related information. The availability of artifacts depends on the

browser, its version, user settings, and whether the records have been deleted or overwritten.

Result:

Thus, browser artifact extraction and examination were studied using browser history, download records, cookies, cache files, and other available browser data. The importance of browser artifacts in reconstructing digital activity during forensic investigations was understood.

EXP NO: 07

COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

Aim:

To compare two files using Compare It software and identify similarities, differences, and modifications in their contents for digital forensic investigation.

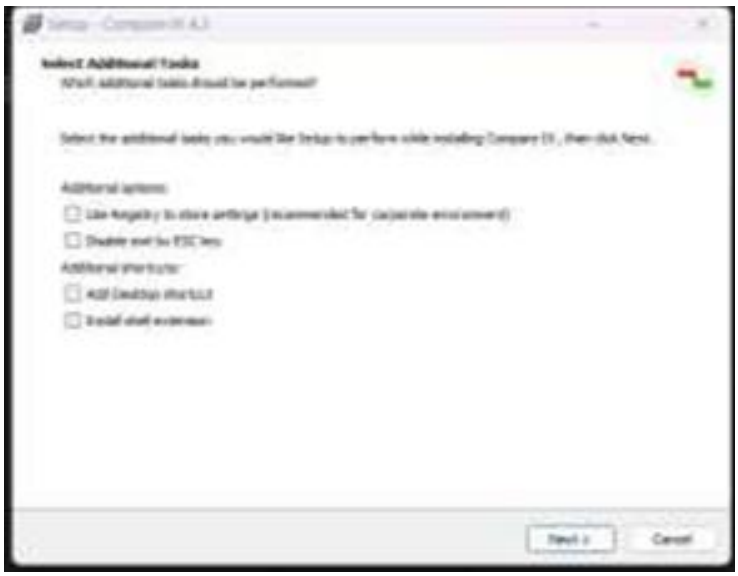
Procedure:

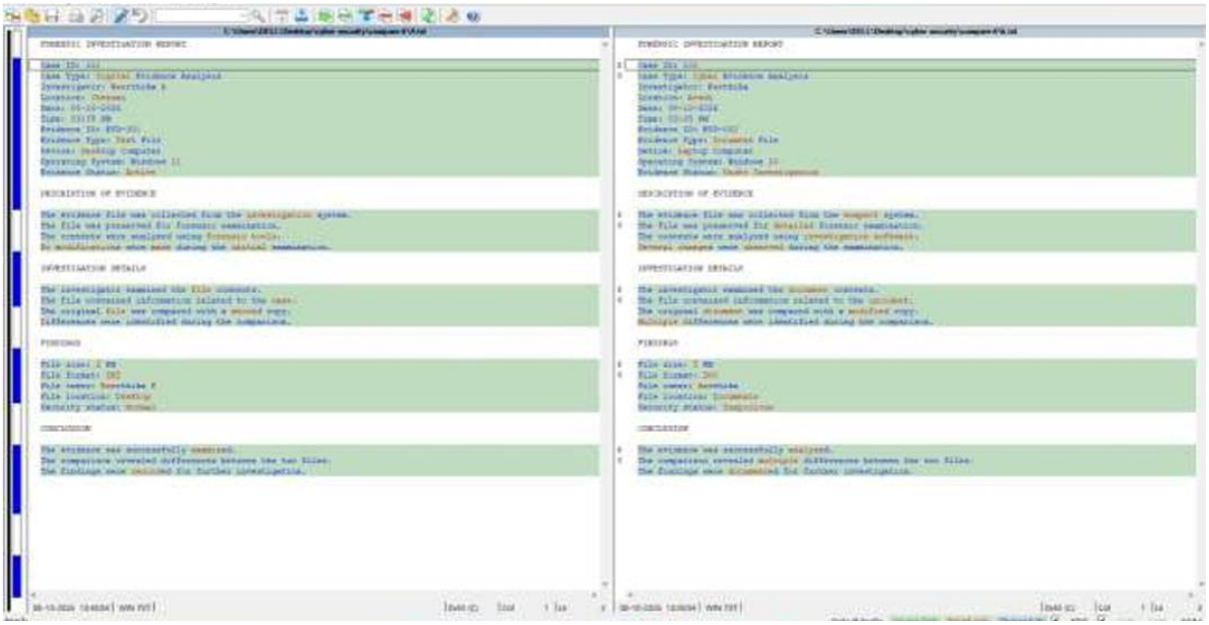
1. Start the computer and launch the **Compare It** software.
2. Prepare the two files that need to be compared. These files may contain text documents, source code, configuration files, log files, or any other supported file type.
3. Store both files in an easily accessible folder and verify their names, file paths, and formats before starting the comparison.
4. Open **Compare It** and create a new comparison session.
5. Select the first file as the **original** or **reference** file.
6. Select the second file as the **comparison** or **modified** file.
7. Click the **Compare** option to begin analyzing the contents of both files.
8. Review the comparison results displayed by the software and identify highlighted differences, including inserted, deleted, and modified lines or sections.
9. Examine the matching portions of both files to identify the content that remains unchanged.

10. Navigate through the detected differences using the software's comparison tools and verify each modification carefully.
11. Save the comparison results, export the comparison report, or capture screenshots of the highlighted differences if the software provides these options.
12. Record the identified similarities, differences, and other relevant observations as part of the forensic investigation documentation.

Observation:

Compare It software helps identify differences between two files by displaying their contents side by side and highlighting changes. This is useful for examining whether a file has been modified.





The comparison may reveal changes in text, lines, or other supported content. However, a visual comparison alone does not establish when a file was modified or who modified it. Hash comparison can help determine whether two files have identical contents.

Result:

Thus, two files were compared using Compare It software, and the similarities and differences in their contents were examined. The use of file comparison in identifying possible modifications during digital forensic investigations was understood.

8.Collection of Email Evidence from a Victim's PC

Aim:

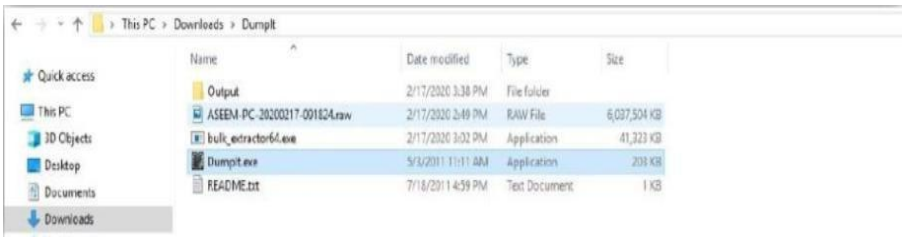
To study the methods used to identify, collect, preserve, and examine email evidence from a computer for digital forensic investigation.

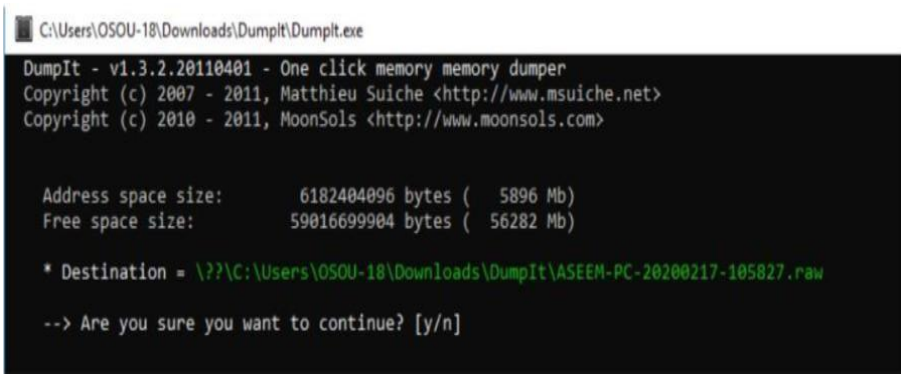
Procedure:

1. Start the computer and obtain the necessary authorization before accessing or examining the email evidence.
2. Identify the email service or email client used on the system, such as Gmail, Microsoft Outlook, or Mozilla Thunderbird.
3. Record important system information, including the date, time, user account details, computer name, and other relevant case information.
4. If the email account is accessed through a web browser, examine the available email messages and collect relevant browser artifacts by following the approved forensic evidence-handling procedures.
5. If a desktop email client is used, locate its local email storage files. For example, Microsoft Outlook typically stores emails in .pst or .ost files, while Mozilla Thunderbird commonly stores messages in MBOX files or other profile directories.
6. Acquire a forensic image of the storage device or create an authorized forensic copy of the relevant email data using a suitable forensic acquisition tool.
7. Preserve the original evidence throughout the investigation and perform all examinations on a verified forensic copy whenever possible to maintain evidence integrity.

8. Export relevant email messages in supported formats, such as .eml or .msg, while preserving their metadata and message headers.
9. Examine each email carefully by reviewing the sender, recipient, subject, date, time, message body, attachments, and available email headers.
10. Analyze all email attachments and record their filenames, file types, sizes, and cryptographic hash values wherever applicable.
11. Calculate cryptographic hash values, such as SHA-256, for the collected evidence files and maintain a detailed chain of custody and evidence log.
12. Document the evidence collection process, forensic tools used, examination steps, timestamps, observations, and final findings in the forensic investigation report.

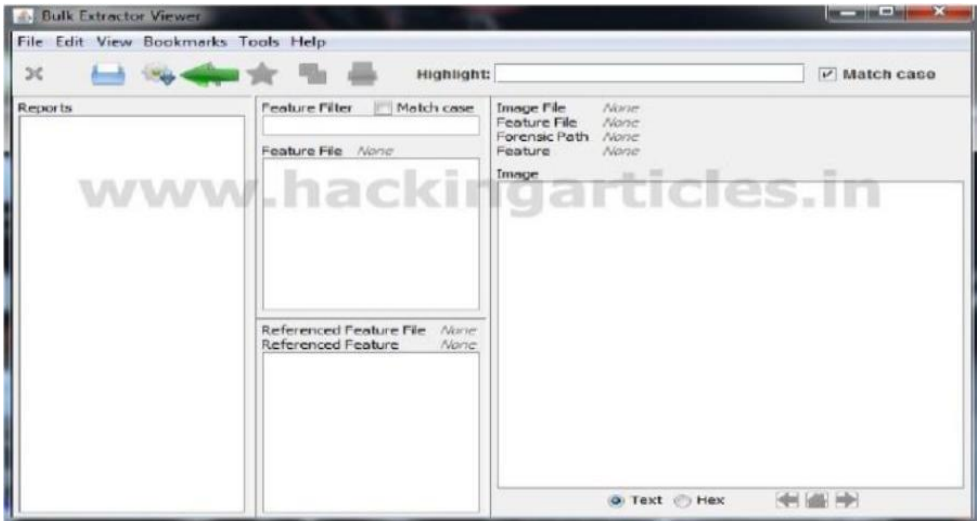
Output:





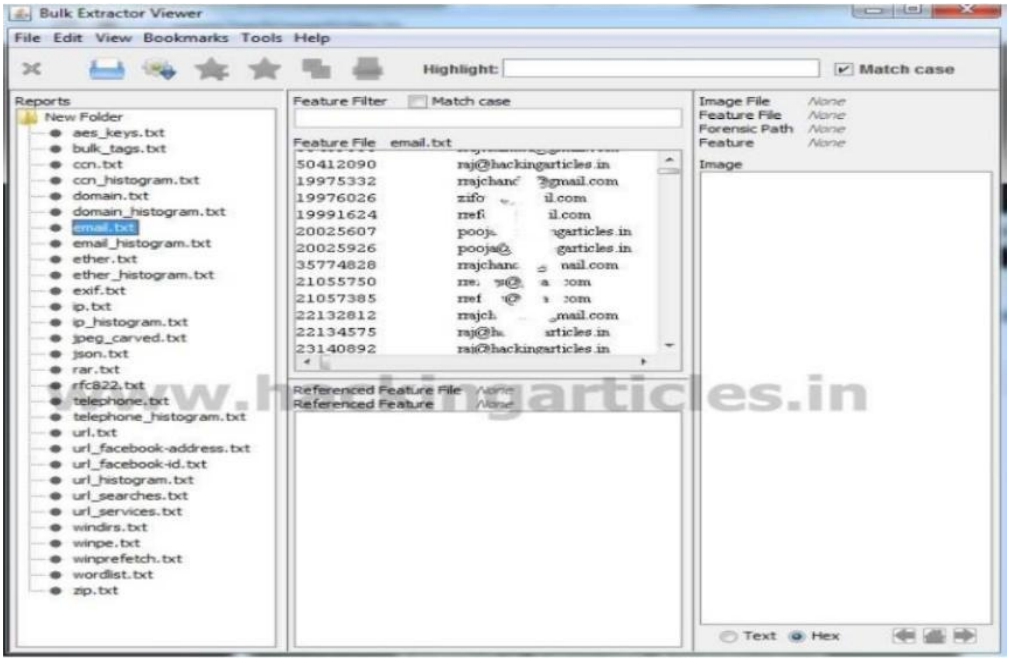
Name: NAREN S Roll No: 241901066

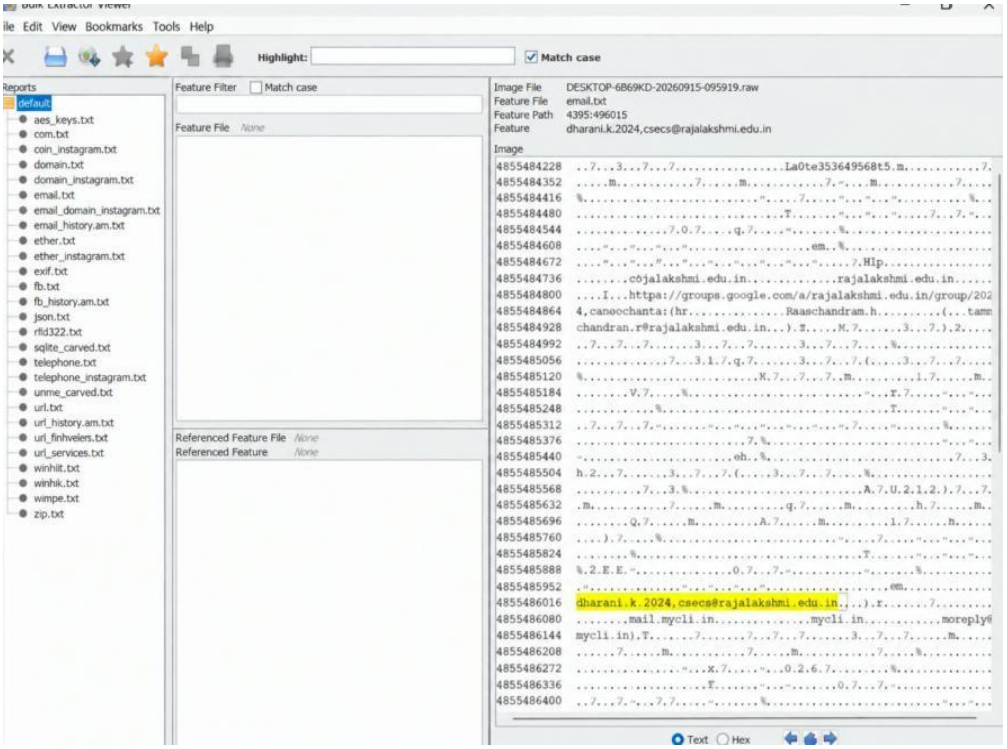
Name	Date modified	Type	Size
Output	2/17/2020 3:38 PM	File folder	
ASEEM-PC-20200217-105827.raw	2/17/2020 4:29 PM	RAW File	6,037,504 KB
bulk_extractor64.exe	2/17/2020 3:02 PM	Application	41,323 KB
DumpIt.exe	5/3/2011 11:11 AM	Application	203 KB
README.txt	7/18/2011 4:59 PM	Text Document	1 KB





Name: NAREN S Roll No: 241901066





Observation:

Email evidence may contain important information about phishing, fraud, unauthorized access, harassment, and other cyber incidents. The evidence must be collected carefully to preserve its integrity.

S. No.	Evidence Type	Information Obtained
1	Sender and Recipient	Email addresses involved in communication
2	Subject	Purpose or topic of the email
3	Date and Time	Available message timestamps
4	Email Header	Routing details and technical metadata
5	Message Body	Content of the communication
6	Attachments	Files shared through the email
7	Local Mail Files	Stored email data from the email client
8	Hash Value	Used to verify whether collected files have changed
9	Evidence Log	Details of collection and handling

Email evidence may be available in a mail server, webmail account, local email client, browser artifacts, or backups. The exact information depends on the email service and the data retained by the system.

Result:

Thus, the methods for collecting and preserving email evidence from a computer were studied. The importance of examining email headers, message content, attachments, and metadata while maintaining evidence integrity was understood.

Name: NAREN S Roll No:241901066

EXP.NO:09

HIDING AND EXTRACTING TEXT FILES BEHIND AN IMAGE USING COMMAND PROMPT

Aim:

To hide a text file behind an image file and extract the hidden text file using Windows Command Prompt as a basic digital forensic experiment.

Part A: Hiding a Text File Behind an Image

1. Create a folder named Forensics on the Desktop.
2. Copy an image file named image.jpg into the Forensics folder.
3. Create a text file named secret.txt in the same folder and enter sample confidential text to be hidden.
4. Open Command Prompt by pressing Windows + R, typing cmd, and pressing Enter.
5. Navigate to the Forensics folder by entering the following command:

```
cd %USERPROFILE%\Desktop\Forensics
```

6. Combine the image and text files using the following command:

```
copy /b image.jpg + secret.txt hidden.jpg
```

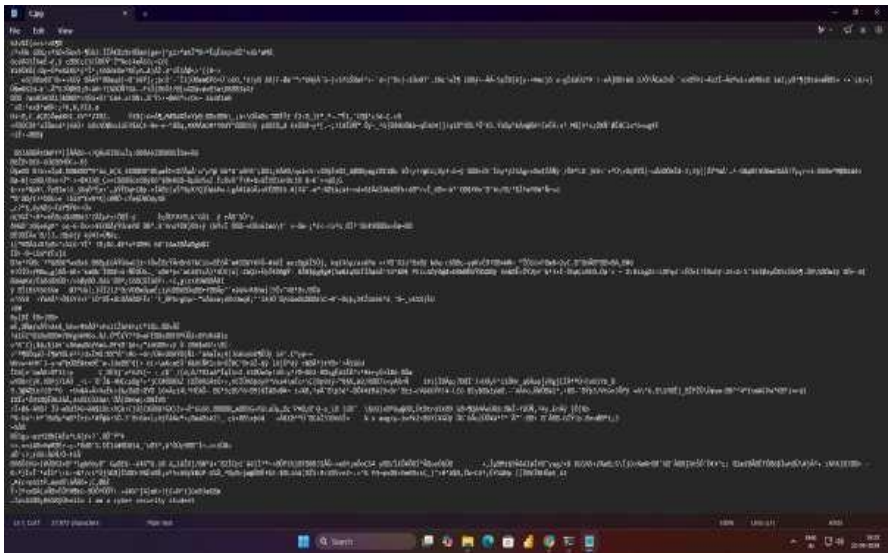
7. Verify that a new file named hidden.jpg has been created. This file appears to be a normal image but also contains the contents of the appended text file.

Part B: Examining and Recovering the Hidden Text

8. Open the newly created hidden.jpg file and confirm that it displays as a normal image without revealing the hidden text during normal viewing.
9. Use a suitable forensic tool or a binary-aware extraction utility to examine the file and locate the appended text data stored after the image content.
10. For demonstration purposes, open hidden.jpg using a text editor and inspect the end of the file. If the appended data is plain text, it may be visible, although this is not considered a reliable forensic extraction method.
11. Recover the hidden text using an appropriate extraction technique and compare it with the original secret.txt file to verify that the hidden information has been successfully preserved.
12. Record the commands used, extracted data, observations, and results of the experiment for forensic documentation.

Note: The copy /b command appends the text file to the image; it does not encrypt the text or conceal it securely. The resulting file may still open as an image, depending on the image format and viewer.

Output:



Result:

Thus, a text file was appended to an image file using Windows Command Prompt. The resulting file was examined to understand how extra data can be stored after image data and how such appended content may be detected during digital forensic investigation.

EXP: 10 Extraction of EXIF Data from an Image File Using EXIF Reader Software

Aim:

To extract and analyze Exchangeable Image File Format (EXIF) metadata from an image using EXIF Reader software for digital forensic investigation.

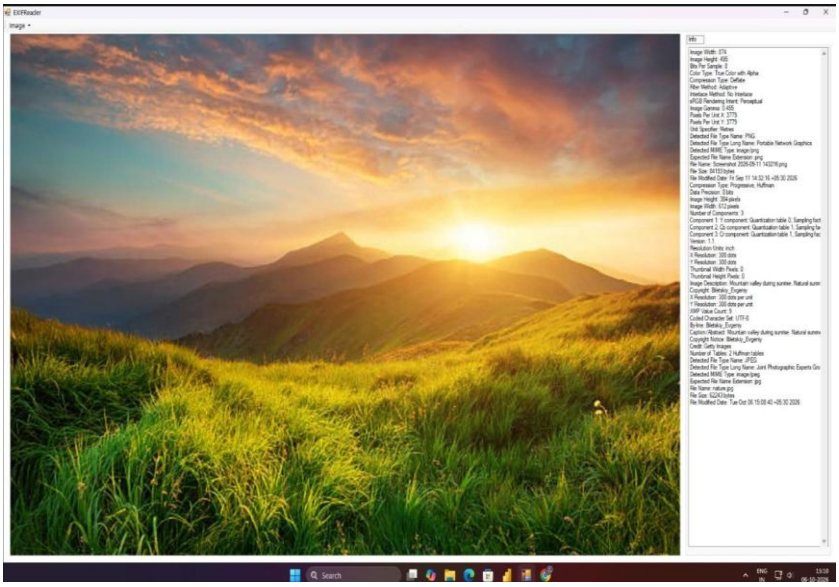
Procedure:

1. Start the computer and locate a suitable image file, such as a JPEG photograph, for forensic examination.
2. Install EXIF Reader software if it is not already available, or launch the application.
3. Open the software and select the option to browse for or open an image file.
4. Navigate to the folder containing the required image and select the file for analysis.
5. Load the selected image into the EXIF Reader application and allow it to process the file.
6. Wait for the software to extract and display the available EXIF metadata associated with the image.
7. Examine important metadata fields, including the camera manufacturer, camera model, image resolution, date and time of capture, orientation, exposure settings, and other available image properties.
8. Check whether GPS coordinates, location information, software details, or device information are available in the metadata.
9. Record the extracted metadata values, including timestamps and other relevant details, in the observation table.

10. If the software provides the option, export or save the metadata report for future reference and forensic documentation.
11. Verify the extracted information and close the EXIF Reader software after completing the examination.

Observation:

EXIF metadata consists of information automatically embedded within an image file by a digital camera, smartphone, or image editing software. It provides valuable details about the image, including device information, capture date and time, camera settings, image dimensions, and, in some cases, GPS location data. During forensic investigations, EXIF metadata helps investigators verify image authenticity, determine when and where a photograph was taken, and support digital evidence analysis.



The following information may be obtained using EXIF Reader:

. No.	Metadata Field	Information Obtained
1	File Name	Name of the examined image
2	File Format	JPEG, TIFF, or another supported format
3	Camera Manufacturer	Camera or device manufacturer, if available
4	Camera Model	Model of the device, if recorded
5	Date and Time	Recorded image capture date and time, if available
6	Image Dimensions	Width and height in pixels
7	Orientation	Orientation information recorded in the metadata
8	GPS Information	Latitude and longitude, if present
9	Software Information	Software used to process the image, if recorded

The availability of EXIF data depends on the image format, device, and processing history. Some applications remove metadata when an image is edited or shared. Recorded timestamps may also be inaccurate or affected by incorrect device settings.

Result

Thus, EXIF metadata was extracted and examined from an image file using EXIF Reader software. The importance of image metadata, including device details, timestamps, image properties, and possible location information, in digital forensic investigations was understood.